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RNN-DAS: A New Deep Learning Approach for Detection and Real-Time Monitoring of Volcano-Tectonic Events Using Distributed Acoustic Sensing

J. Fernández-Carabantes[1,2], M. Titos[3], L. D'Auria[4], J. García[3], L. García[3], and C. Benítez[3]

[1]Departamento de Física Teórica y del Cosmos, Universidad de Granada, Granada, Spain, [2]Instituto Andaluz de Geofísica, Universidad de Granada, Granada, Spain, [3]Departamento de Teoría de la Señal, Telemática y Comunicaciones, Universidad de Granada, Granada, Spain, [4]Instituto Volcanológico de Canarias (INVOLCAN), Puerto de la Cruz, Spain

Key Points:

- RNN-DAS is a deep learning model using LSTM-based RNNs for real-time volcano-seismic signal recognition with DAS data
- Able to detect events with over 97% accuracy by analyzing the evolution of signal energy across frequency bands
- It achieves strong generalization, high interpretability, fast real-time monitoring, and low computational load

Correspondence to:

J. Fernández-Carabantes,
javierfyc@ugr.es

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Author Contributions:

Data curation: L. D'Auria
Investigation: J. Fernández-Carabantes, M. Titos
Methodology: J. Fernández-Carabantes, M. Titos, C. Benítez
Project administration: C. Benítez
Software: J. Fernández-Carabantes, M. Titos, J. García, L. García, C. Benítez
Supervision: C. Benítez
Validation: M. Titos, L. D'Auria
Writing – original draft: J. Fernández-Carabantes
Writing – review & editing: M. Titos, L. D'Auria, J. García, L. García, C. Benítez

Abstract

We present a novel Deep Learning model based on recurrent neural networks (RNNs) with long short-term memory (LSTM) cells, designed as a real-time volcano-seismic signal recognition system for distributed acoustic sensing (DAS) measurements. The model was trained on an extensive database of volcano-tectonic events derived from the co-eruptive seismicity of the 2021 La Palma eruption, recorded by a High-fidelity submarine distributed acoustic sensing array near the eruption site. The features used for supervised model training, based on the average signal energy in frequency bands, enable the model to effectively leverage the spatio-temporal contextual information of seismo-volcanic signals provided by the technique. The proposed model not only detects the presence of volcano-tectonic events but also analyzes their temporal evolution, selecting and classifying their complete waveforms with an accuracy of approximately 97%. Furthermore, the model has demonstrated robust performance in generalizing to other time intervals and volcanoes. Such results highlight the potential of using RNN-based approaches with LSTM cells for DAS systems located in volcanic regions, enabling fast, automatic analysis with low computational requirements and minimal retraining. This allows continuous real-time monitoring of seismicity while facilitating the creation of labeled seismic catalogs directly from DAS measurements, representing a significant advancement in using DAS technology as a viable tool to study active volcanoes and their seismic activity.

Plain Language Summary

Among the existing geohazards, volcanic eruptions are particularly significant due to their impact on nearby populations, including their related seismic activity and gas emissions. Traditionally, scientists studying them have employed seismic networks in volcanic environments to obtain rich insights about the state and evolution of volcanic systems. However, the deployment and maintenance of these monitoring networks are costly and complex in most cases. The introduction of an innovative technique called distributed acoustic sensing (DAS) allows the use of fiber optic cables as arrays of thousands of sensors, covering large areas and providing a valuable tool for volcano monitoring. However, due to its characteristics, the volume of data it generates is unmanageable manually by experts. To address this challenge, we have developed the RNN-DAS model using advanced deep learning techniques. Which has proven to be capable of analyzing DAS seismicity in real time. And thus, it can serve as a support tool for scientists, enabling the use of DAS as a viable technique for studying active volcanoes without the need for constant and tedious human oversight.

1. Introduction

The application of seismology to the study of volcanic phenomena has proven to be one of the most effective tools for obtaining valuable geophysical information about the dynamic behavior of volcanoes and understanding their eruptions (Iyer, 1992). This success is attributed to the diverse origins of seismic signals generated by magma transport, changes in hydrothermal regimes, and pressure variations within the volcanic edifice (B. Chouet, 2003; B. A. Chouet & Matoza, 2013). Among the different types of events, those known as Volcano-Tectonic (VT) events are the most similar to traditional tectonic earthquakes, sharing the same source mechanism and exhibiting P and S phases. These events are characterized by high frequencies and are of significant interest for monitoring volcanic activity (Roman et al., 2016; White & McCausland, 2016). VT events are caused by the fracturing of volcanic materials or the rupturing of nearby faults in response to dynamic changes within the volcanic system. Consequently, they can serve as precursors to a volcanic eruption in the form of seismic swarms, highlighting the importance of their study (Afnimar et al., 2022). However, their recordings can differ from classical tectonic events due to the high heterogeneity of the volcanic medium through which they propagate (Almendros et al., 2007; Lahr et al., 1994; Zobin, 2003).

Detection and localization of volcano-seismic events have traditionally required a large spatial deployment of broadband seismic stations in the volcanic environment (B. Chouet et al., 1997; Dreier et al., 1994), either as dense networks or seismic arrays (Almendros et al., 2002; Ibáñez et al., 1997). However, such configurations are expensive and challenging to maintain over the long term, being influenced by meteorological noise or ground tilt. As a result, most volcanoes rely on limited permanent station deployments, which may be insufficient to fully monitor volcanic unrest (De Barros et al., 2013; Wassermann et al., 2022).

In this context, promising new methodologies have been developed in recent years within the field of acoustics to address the aforementioned deployment challenges. Of particular interest is the Distributed Acoustic Sensing (DAS) technique (Zhan, 2020). By measuring deformations along an optical fiber cable, using laser pulse emissions through an interrogator and analyzing Rayleigh scattering, produced at imperfections in the fiber caused by wavefront propagation (Hartog, 2017; Lindsey et al., 2020), it is possible to perform seismic event measurements in an array-like configuration of a single channel with unprecedented spatial and temporal coverage (Fichtner et al., 2023; Williams et al., 2019). This approach enables a logistically viable deployment across diverse environments, including glaciers (Walter et al., 2020), the seafloor (Lior et al., 2021; Nakano et al., 2024), and highly heterogeneous and topographically complex settings such as volcanic terrains (Biagioli et al., 2024; Klaasen et al., 2021). Furthermore, DAS provides broadband response capabilities (Lindsey et al., 2020), with strain responses comparable to those of traditional point-like stations in volcanic environments (Currenti et al., 2021), high temporal sampling frequencies (up to kHz), spatial resolutions on the order of meters, and measurement distances on the order of kilometers with only one interrogator (Lindsey et al., 2017). These attributes explain the growing interest in applying DAS to volcanic seismology, with promising results demonstrated in locating tremor and explosion events using array techniques at Stromboli (Biagioli et al., 2024), hypocenter localization and site response analysis at Azuma volcano (Nishimura et al., 2021), very long period events at Vulcano (Currenti et al., 2023), degassing-related events (Jousset et al., 2022), submarine volcanic-related observations (Nakano et al., 2024), dike intrusions (Li et al., 2024), characterization of volcanic calderas (Biondi et al., 2023) or structural imaging (Jousset et al., 2018).

One of the most complex objectives in volcanic seismology is the development of robust databases of well-classified events using recognition pattern mechanisms to characterize them. Traditionally, the methodology employed for event detection and classification involved manual picking performed by expert geophysicists with years of experience (Ibáñez et al., 2000). This approach, while effective, is extremely time-consuming and further complicated by the vast amount of data generated by the high spatiotemporal sampling of DAS. In seismo-volcanic applications, this poses significant challenges for analyzing the recorded seismicity, both in terms of human and computational resources. This challenge marked the initial point where the use of Machine Learning techniques became important and proved to be game-changers. Their use has expanded and demonstrated efficacy in traditional volcanic seismology for detecting and classifying various types of events through different models: Support Vector Machines (R. Lara-Cueva et al., 2017; Masotti et al., 2006), Decision Trees (R. A. Lara-Cueva et al., 2016), and Hidden Markov Models (Beyreuther & Wassermann, 2008; Ibáñez et al., 2009). More recently, neural network models have facilitated the application of Deep Learning techniques to the classification of seismo-volcanic events, yielding promising results (Esposito et al., 2018; Malfante et al., 2018; Titos, Bueno, García, & Benítez, 2018). Convolutional Neural Networks (CNNs), based on filter operations to extract signal features, have also been successfully explored (Canario et al., 2020; Lara et al., 2021; Titos et al., 2019). Additionally, Recurrent Neural Networks (RNNs), capable of leveraging the temporal evolution of seismo-volcanic signals for classification, have shown successful implementation (Titos, Bueno, García, Benítez, & Ibañez, 2018).

Despite the success of Deep Learning in detecting and classifying volcanic events from traditional seismic recordings, the same cannot be said for its application to DAS. To date, no specific models exist for detecting and classifying VT events in DAS near volcanoes. This limitation is attributed to the novelty of the technique and the complexity of managing the large volumes of data involved, rather than a lack of interest (Biondi et al., 2024). However, there are models adapted from traditional seismology to DAS that have shown promising results in event detection. Notable examples include the CNN-based model PhaseNet-DAS (Zhu et al., 2023a) and a transformer-based model (Corsaro et al., 2024), which are focused on automatic picking. This gap has led some researchers to rely on manual picking (Nishimura et al., 2021) or well-established methods in traditional seismology, such as the STA/LTA approach (Allen, 1978; Biagioli et al., 2024; Jousset et al., 2022; Klaasen et al., 2021; Nakano et al., 2024), when studying volcanic related seismicity with DAS. While these methods can be effective, they require carefully tuned parameters (Trnkoczy, 2009) and are negatively affected by low Signal-to-Noise Ratios (SNR). A common issue with DAS as data tend to be noisier than traditional seismic measurements, with additional challenges posed by instrumental noise and signal attenuation due to the epicentral distance of the event to the DAS array (Baba et al., 2024; Zhu et al., 2023a).

For this reason, in this article, we present a Deep Learning model (RNN-DAS) based on recurrent neural networks with Long Short-Term Memory (LSTM) cells (LeCun et al., 2015). The model was trained in a supervised fashion using an automated database of events detected by an underwater high-fidelity DAS array deployed 10 km appart from the site of the destructive Tajogaite volcano during the eruption in 2021 on the island of La Palma, Spain (Carracedo et al., 2022; Troll et al., 2024). This model has been specifically designed to function as a real time Volcano-seismic Signal Recognition (VSR) system. It leverages features that allow the network to utilize the high spatial and temporal information provided by DAS without requiring adjustments for site effects or low SNR. Unlike previous models, the proposed one not only detects events but is also capable of selecting the waveform of the event, providing precise identification and extraction of the signals of interest. The methodology adapts the approach proposed by (Titos, Bueno, García, Benítez, & Ibañez, 2018) to enable the detection, classification, and real-time monitoring of events based on signal energy within frequency bands. This system has demonstrated excellent performance, achieving results comparable to those of visual detection by experts using conventional seismic stations.

As an increasing number of people are exposed to the dangers of volcanic eruptions (Small & Naumann, 2001), the relevance of methodologies like this system grows significantly. These systems enable the automatic analysis and creation of volcano-seismic signal databases recorded by early warning systems in near-real time. Such capabilities allow geophysicists to evaluate the evolution of volcanic activity, facilitating the implementation of civil protection plans to mitigate material damage and save human lives (Brown et al., 2017; Berz et al., 2001; Hewitt, 2014; R. A. Lara-Cueva et al., 2016).

The rest of this paper is structured as follows. Section 2 provides a brief explanation of the fundamentals of the DAS technique. Section 3 focuses on the capability of deep learning architectures to monitor continuous signals, justifying the use of RNNs with LSTM units to address the challenge of modeling the temporal evolution of seismic signals recorded by DAS. Section 4 describes the experiment conducted, including the characteristics of the HDAS used, the methodology for recording and selecting seismic events, the approach for extracting relevant features, the automatic labeling process for supervised model training and the generalization tests implemented to study the model capabilities. Section 5 presents the results and discussions of the model performance. Finally, Section 6 concludes the study.

2. Distributed Acoustic Sensing (DAS) Technique

In this section, the fundamentals of the Distributed Acoustic Sensing (DAS) technique are briefly presented. For a detailed explanation of its methodology and the underlying physical-mathematical principles, readers are encouraged to consult the works by Lindsey et al. (2020), Zhan (2020), and Shang et al. (2022).

DAS is a novel technology recently applied in seismology, enabling continuous monitoring of strain (ε) or strain rate ($\dot{\varepsilon}$) along a fiber optic cable when an acoustic wavefront (such as the one produced by seismic elastic waves from an earthquake) passes through it over distances of tens of kilometers. DAS offers ultra-dense spatiotemporal sampling at a low cost and with efficient deployment. Apart from the optic fiber cable, the other key element is the DAS interrogator. Which is positioned at the origin of the fiber optic cable, and continuously emits pulses of laser light. When a wavefront encounters the fiber and induces perturbations, natural imperfections within the cable cause Rayleigh scattering of the light pulses. A portion of this scattered light is retro-dispersed back to the interrogator. By measuring the phase shift of these retro-dispersed pulses, the strain or strain rate is calculated using interferometric techniques, with the imperfections serving as strain sensors distributed along the fiber.

This process enables probing over distances of up to tens of kilometers along the fiber, with temporal sampling frequencies reaching the kHz range and spatial sampling at the meter level. As a result, the DAS measurement can

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be interpreted as that of a strainmeter array, offering broadband characteristics, high fidelity of waveforms, and a good strain transfer response from the ground to the cable, making it suitable for seismological studies. The measurement with a DAS array depends on various instrumental and deployment parameters. While different models exist depending on the manufacturer, in most cases, a single DAS interrogator can probe distances of up to 50 km. The characteristics of the emitted laser pulses, such as their emission rate, width, and power, are also crucial. DAS provides distributed measurements, meaning that the spatial points or DAS channels where strain or strain rate is measured are located at the midpoint of what is known as the Gauge length, which is the spatial interval over which strain is computed (typically on the order of tens of meters). The Gauge length is related to the width of the laser pulse, and the distance between DAS channels, defined by the step between Gauge lengths, determines the spatial sampling interval (dx). Another relevant factor is the temporal sampling frequency (dt) and the time over which strain is derived along the channels, known as derivation time, used for computing the strain rate.

In addition to these advantages, the use of DAS interrogators on pre-existing optical fibers, commonly referred to as *dark fibers*, offers an easy and cost-effective application method. However, there are certain limitations inherent to the technique that present challenges. First, if the full potential of the technique is exploited, the volume of data generated can be immense, reaching terabytes per day. Further challenges include potential site effects, coupling effects with the ground, meteorological drifts in the measurements, and the precise location of the DAS channels or fiber cable.

3. Deep Architectures for Continuous Monitoring

Deep learning methods, originating from the field of artificial neural networks (ANNs), offer significant potential for addressing complex challenges in continuous signal processing. Among these, Recurrent Neural Networks (RNNs) have been widely used for modeling sequential data, particularly with the incorporation of Long Short-Term Memory (LSTM) networks.

Given their ability to capture temporal dependencies, RNNs have been a natural first choice for initial experimentation in our study. Their sequential nature allows them to process and store memories of arbitrary input sequences, making them particularly well-suited for analyzing continuous data streams, such as detecting seismic signals (LeCun et al., 2015; Schmidhuber, 2015). Unlike feedforward neural networks, RNNs generate outputs not only from the current input but also from prior inputs through a hidden state that is updated over time. This capability makes them effective for tasks requiring complex temporal modeling (Kamath et al., 2019).

Previous studies have explored the integration of RNNs with Convolutional Neural Networks (CNNs) to enhance feature extraction and temporal modeling in seismic analysis (Canario et al., 2020), as well as the use of other deep learning architectures, such as Transformers, to study seismic signals (Münchmeyer et al., 2021). While more complex architectures could be explored in the future, we focus on RNNs as an initial step due to their simplicity, lower computational requirements, reduced number of parameters compared to more complex architectures, and greater explainability in neural activation patterns, which make them well-suited for our objectives (Titos et al., 2024).

However, training RNNs can be challenging due to vanishing or exploding gradients, limiting their ability to model long-term dependencies (Pascanu et al., 2013). LSTM networks (Hochreiter & Schmidhuber, 1997) mitigate these challenges by incorporating specialized gates that regulate the flow of information, enabling efficient storage, updating, and forgetting of information over extended sequences (Kamath et al., 2019).

RNNs have been successfully applied across various scientific fields, including speech recognition (Graves et al., 2013), medical diagnostics (Choi et al., 2017), meteorological forecasting (Balluff et al., 2020), image generation (Gregor et al., 2015), and genetics (Xu et al., 2007). Furthermore, they have proven effective in geophysical applications such as volcano monitoring (Titos, Bueno, García, Benítez, & Ibañez, 2018). Given these strengths, RNNs with LSTM remain a compelling choice for our initial approach in capturing the temporal evolution of seismic signals recorded by the DAS array while maintaining high model explainability and simplicity.

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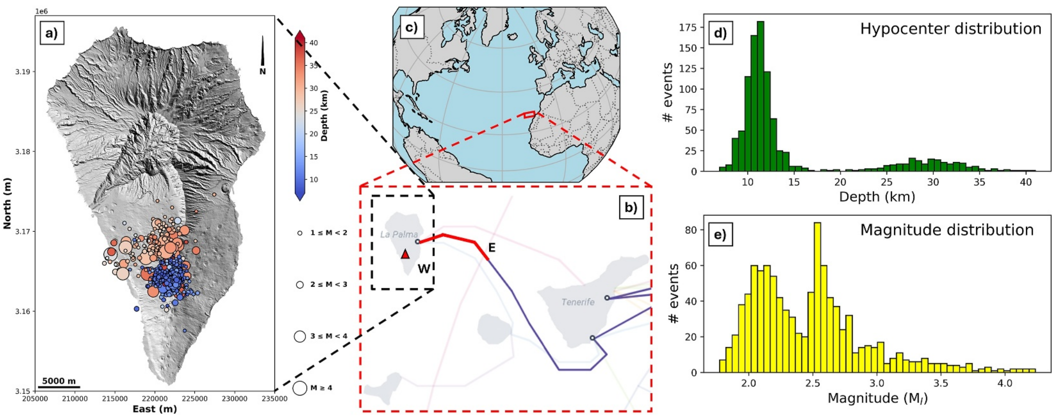

Figure 1. (a) Map of La Palma showing the VT events used to train the model, where color indicates the depth of the hypocenter and the size of the epicentral circle represents the magnitude. The digital elevation model was obtained from Copernicus (2022). (b) Map of the portion of the submarine cable (red line) used for the HDAS array. Data obtained from Barrancos et al. (2022). (c) Location map of the Canary Islands, Spain. (d) Histogram of the hypocenter distribution for the events present in the model database used for RNN-DAS. (e) Histogram of the magnitude distribution for the events present in the database used for RNN-DAS.

4. Experimental Design

4.1. Description of the Field Experiment: La Palma 2021 Eruption

The 2021 La Palma eruption occurred on the volcanic ridge of Cumbre Vieja, the most active area from a volcanological perspective of the on the island of La Palma, Canary Islands, Spain. The eruption, which lasted from 19 September to 13 December 2021, was characterized as the longest in the historical record of the island, the one with the highest amount of volcanic material ejected, and also the most destructive (Carracedo et al., 2022; Del Fresno et al., 2023).

The eruption took place along a series of fissural openings with the emission of basaltic lavas, generating a cinder cone in the area of greatest effusive activity, known as the Tajogaite or Cabeza de Vaca volcano (Carracedo et al., 2022). The eruption was marked by different periods of explosivity (Romero et al., 2022) and was preceded by significant seismic activity. Since 2017, the seismic reactivation of the island had been evident, with swarms of VT events and uplift deformation indicating signs of major disruptions in the internal system of Cumbre Vieja (Torres-González et al., 2020). Finally, a week prior to the eruption, intense seismic activity of VT events occurred, associated with pressure changes in the internal systems of Cumbre Vieja due to magma ascent and migration toward the eruptive vents (D'Auria et al., 2022). Co-eruptive seismicity persisted in the form of different types of volcano-seismic events, including low-frequency volcanic tremor, long-period events, and VT events. The latter occurred in two depth clusters with a notable gap between them (see Figure 1a), driven by aseismic magma transport and consequent dynamical changes between shallow and deep reservoirs (Del Fresno et al., 2023).

During the eruption, the INstituto VOLcanológico de CANarias (INVOLCAN) and its partners installed, on October 19, a High-fidelity DAS (HDAS) system, provided by the company Aragon Photonics (https://aragon-photonics.com/products/hdas/), using a submarine fiber-optic cable that connects the islands of La Palma and Tenerife, operated by the company Canalink, starting near the coast close to Santa Cruz de La Palma at approximately 28.663°N, −17.765°E (see Figures 1b and 1c). This HDAS, installed approximately 10 km from the Tajogaite volcano, recorded all co-eruptive seismic activity, including tremor, VT events, regional tectonic events, and teleseismic events. After the eruption, it continued recording measurements and remains operational to this day. The HDAS recorded strain rate data at a sampling frequency of 100 Hz, with a spatial sampling interval of 10 m, gauge length of 10 m, and an initial total distance of 30 km, later extended to 50 km via Raman amplification. Initially, only 2,976 DAS channels were available, later, when the probing length was increased, a total of 4,960 DAS channels were employed (Barrancos et al., 2022).

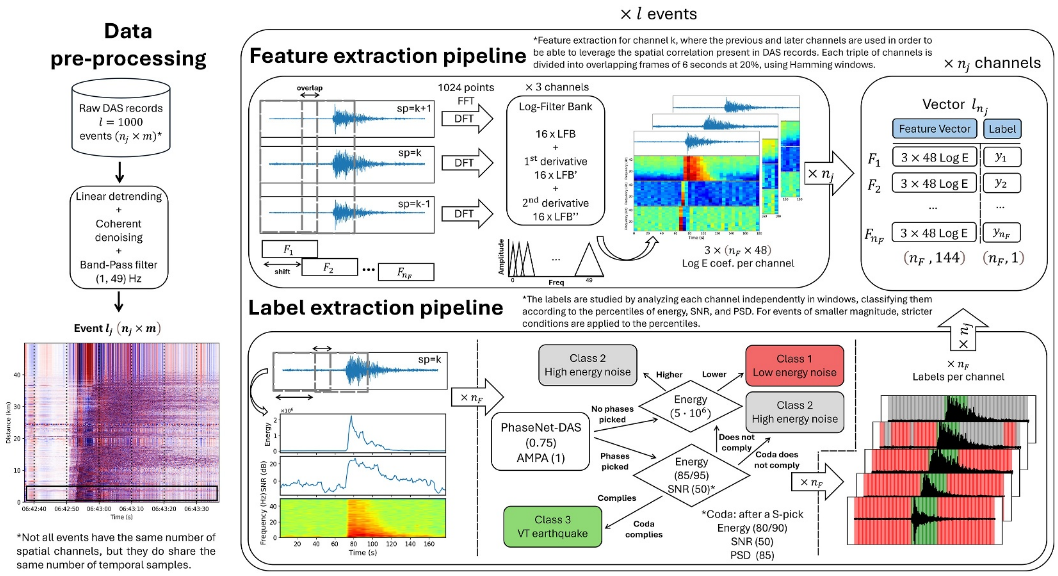

Figure 2. Schematic of the methodology employed in the preprocessing pipelines for DAS data, feature extraction representing energy coefficients that account for the temporal evolution and spatial coherence of the studied signals, and the label extraction pipeline for supervised training (with the threshold percentile values applied to each variable, depending on whether the event is of high or low magnitude). This procedure is followed for each of the n_j channels of each event l in the data set, obtaining the feature and label values for each frame F_k of the signals.

4.2. Database Description: Pre-Processing, Features, and Labels Extraction Pipelines

Using the seismic catalog of events recorded during and after the eruption by INVOLCAN, a total of 1,000 VT events recorded by the HDAS during November 2021 were selected to train the RNN-DAS model. These events exhibit a bimodal depth distribution (Del Fresno et al., 2023) and a magnitude distribution characterized predominantly by small to moderate magnitudes (see Figures 1d and 1e). Several key aspects must be considered when training the model:

1. The presence of natural, anthropogenic, or internal noise can significantly affect the recordings of the events, particularly in HDAS array data, which are inherently influenced by low SNR conditions. This leads to a degradation of seismic signal quality (Bormann & Wielandt, 2013).
2. Site effects in the measurements, as the seafloor conditions where the HDAS is deployed influence how the events are recorded and their coupling with the ocean floor (Lior et al., 2021).
3. Seismic waves are affected during their propagation by the medium through a series of effects that alter their recording. These effects are amplified in highly heterogeneous volcanic environments due to their abrupt topography and lateral velocity discontinuities, leading to attenuation, scattering, diffraction, and path deviation of the seismic energy (Almendros et al., 2001; Durek & Ekström, 1996; Schwarz, 2019).

The procedure for data pre-processing and the extraction of labels and features for model training is explained in detail below, being a modified version of the approaches proposed by Titos, Bueno, García, Benítez, and Ibañez (2018) and Benítez et al. (2006) (see Figure 2).

1. *Data pre-processing pipeline*. For each event, based on its origin time from the catalog, 3 min of HDAS signal are selected to ensure the complete signal is captured by including 1 min before and 1 min after the event. First, a linear detrending is applied to the data, followed by denoising to remove coherent noise, and finally a double 4th-order causal band-pass filter between 1 and 49 Hz is applied, as the focus is on detecting VT events.
2. *Feature extraction pipeline*. To enable the RNN-DAS model to learn meaningful features, we employ a signal parametrization based on Logarithmic Filter Banks (LFB), which facilitate the extraction of a coefficient vector that estimates the signal's energy in frequency bands, offering higher resolution at lower frequencies. This methodology has been recognized as particularly relevant for studying seismo-volcanic signals (Benítez et al., 2006; Ibáñez et al., 2009; Titos, Bueno, García, Benítez, & Ibañez, 2018). Rather than directly extracting features from the temporal domain, this approach is favored due to the inherent variability of seismo-volcanic

events, where both temporal duration and volcanic system behavior can fluctuate significantly. Furthermore, the simultaneous recording of multiple events can complicate the direct modeling of temporal behavior from raw input data. For these reasons, signal parametrization remains a robust and suitable approach.

To obtain the feature vector of the signal associated with channel k, F_k, from the DAS record, we simultaneously use the records from channels $k-1$ and $k+1$. The signal is divided into n_F frames using 6-s Hamming windows with 20% overlap. In the case of 3-min samples used during the training procedure, this results in $n_F = 37$. This approach provides both temporal and spatial overlap of the information. For the spatial overlap, we use groups of 3 channels, consisting of the target channel along with its previous and next channels, creating a spatial overlap between the DAS channels. Then, when dividing the signal into temporal frames, a temporal overlap is applied to the information within each frame across time for the tuple of channels.

This configuration was chosen to strike a balance between the accuracy required for signal detection and the computational load of the model, enabling real-time analysis. For each of the n_F frames, a 1024-point FFT is computed, producing 16 LFB coefficients, 16 coefficients for the first temporal derivative of the LFB vector, and another 16 for the second derivative, for each of the 3 channels. This results in a 144-component vector for each frame:

$$F_k^{n_F} = \left(\mathrm{LFB}_i, \mathrm{LFB}_i', \mathrm{LFB}_i'' \mid i \in \{k-1, k, k+1\}\right) \Rightarrow F_k = (n_F, 144). \tag{1}$$

where in the second part of the equation, the features from all frames of channel k are stacked into a matrix. Once this stacking is performed for all n_j spatial channels of the HDAS, with $n_j = 4960$, the result is a feature matrix for event l with dimensions $F_l = (n_j, n_F, 144)$.

This configuration was found to best model temporal sequences while leveraging the full potential of the spatial and temporal sampling offered by the DAS technique. By incorporating contiguous channels, coherent spatial information is utilized, and by analyzing the LFB coefficients along with their first and second derivatives, the evolution of energy, its velocity, and acceleration across the signal are studied. This feature set has proven to be highly informative for the analysis of DAS signals and, consequently, beneficial for our model as well (Fakhruzi et al., 2025). While using only the LFB coefficients yielded good results for high-SNR events, for lower-magnitude events, low-SNR signals, or highly attenuated signals, the inclusion of the derivatives significantly enhanced the model's ability to highlight energy changes caused by the arrival of seismic phases associated with a VT event, thereby improving the model's identification capability.

3. *Label extraction pipeline*. Due to the large number of events included in the database and the waveform analysis required for each channel, the use of a manually labeled database for supervised training is unfeasible. Instead of manual labeling, the process is performed automatically based on signal parameters. To achieve this, we defined three distinct classes for the classification task: Label 2 corresponds to the presence of an event, while Label 0 and Label 1 represent the absence of events, differentiated as *Noise Type 1* and *Noise Type 2*, respectively. These noise classes are separated by an energy threshold derived from differences observed between quieter and noisier channels. This distinction allows us to differentiate between low-energy natural seismic noise and higher-energy noise, which may originate from external sources such as anthropogenic activities or from DAS channels affected by strong site effects or poor coupling with the seafloor. These channels can show signal amplitudes up to two orders of magnitude greater than neighboring quieter channels. By incorporating this three-class structure, the labeling process helps prevent misclassification of events in noisy channels and enhances the model's robustness in detecting event-related activity throughout the entire fiber optic array.

   Considering that each selected signal sample contains a cataloged event, the class of each frame is determined based on its energy content, power spectral density (PSD), and SNR values relative to a set of thresholds. These thresholds are manually fine-tuned based on signal percentiles, and the class is assigned through a sequential decision process (see Figure 2).

   First, given the possibility that the event may be indistinguishable from background noise due to attenuation, the PhaseNet-DAS (Zhu et al., 2023a) and AMPA models (Álvarez et al., 2013) are applied with thresholds of 0.75 and 1, respectively. These models are used to detect whether any P or S phases are present, or whether no event is detectable. If phases are identified, the frames exceeding the energy and SNR thresholds are labeled as VT, while the remaining frames are classified as noise. Based on their energy levels, the noise frames are further subdivided into the two defined noise classes. To improve the analysis of the coda, if an S-phase is detected, the energy threshold is reduced by 5%, and an additional PSD condition is introduced. A coda frame is assigned to the event class if it exceeds both the 85th percentile PSD threshold and the adjusted energy threshold. If the frame does not exceed the PSD threshold but surpasses the energy threshold, it is classified as *Noise Type 2*. It is worth noting that the thresholds for the metrics are adjusted based on event magnitudes. For events with magnitudes below 2.5, stricter energy thresholds are imposed by increasing the percentile threshold by 10% to avoid mistakenly labeling frames without a signal.

Energy estimation is performed by calculating the zero-lag autocorrelation within each frame, which corresponds to the squared sum of the signal coefficients (Dondurur, 2018). The PSD is estimated frame-by-frame and normalized to a range between 0 and 1 using the minimum and maximum values of each studied signal sample. The SNR is estimated in decibels (dB) for each frame, using a 3-min noise sample from the HDAS as a reference.

As with the feature extraction process, each signal from channel k is divided into n_F frames of equal length via Hamming windowing, and a class label is assigned to each frame. Consequently, the label vector for each channel is represented as $L_k = (n_F, 1)$, and when stacking the labels for the n_j channels in the DAS array, the resulting label matrix has a final dimension of $L_l = (n_j, n_F, 1)$.

These labels were manually reviewed by an expert in several cases to ensure a generally accurate labeling process and to verify the manually fine-tuned set thresholds.

4.3. Design of the RNN-DAS Model

To design the RNN models with LSTM cells, PyTorch (Paszke et al., 2019), a Deep Learning framework implemented in Python, was used. Additionally, all database pre-processing prior to training, as described in Section 4.2, was also implemented in Python version 3.7. The data set used for the training procedure was split into training (65%), validation (17.5%), and test (17.5%), with a random shuffle applied to balance the sets.

The hyperparameters that yielded the best results were as follows: a batch size of 256, 256 hidden states, 1 layer depth for the model, and a variable learning rate [0.1, 0.0001] controlled by a cosine scheduler. To compute training errors, softmax normalization was applied to predictions to obtain class probabilities, and the cross-entropy criterion (Hinton et al., 2012) was used. SGD was employed as the optimizer to adjust the model weights during each training epoch.

Two different models were trained. The first one, referred to as RNN-DAS (1000), was trained for 100 epochs using the database of 1000 events described earlier, with the characteristics specified above. The second model, RNN-DAS (1150), involved fine-tuning the first model with a balanced database of 150 events, consisting of 50 events with magnitudes above 2.5, 50 events with magnitudes below 2.5 (both sets randomly selected from the events used to train the first model), and 50 noise-only samples selected from time periods in which no events were reported in the seismic catalog, thus representing clean background noise. This fine-tuning was performed over 10 epochs, with the learning rate reduced by 10% relative to the first model. The goal of this fine-tuning was to enable the RNN to also model continuous signals without events.

An important consideration is ensuring the model's applicability to DAS records with varying signal amplitudes and energy levels. To address this, the features of all events in the training set were standardized using Z-score normalization, by computing the mean and standard deviation for each of the 144 features used by the model. These statistics were then applied to standardize the validation and test sets, as well as any input data for which the model made predictions, ensuring that all data are processed under the same statistical conditions as the training set.

Given that each HDAS record consists of n_j channels and the event records for individual channels can differ, it was necessary to assign specific labels and features to each channel in the database. As a result, a significantly larger data set of individual seismic waveforms was utilized for training. This process led to the creation of an automated benchmark database comprising over 3 million unique VT waveforms. This ensures robustness against potential outliers during feature and label extraction, resulting in models trained on a large variety of VT event waveforms.

Given the high computational load associated with training our deep learning model and the large amount of DAS data involved, the task was carried out using an NVIDIA GeForce GTX 1080 Graphics Processing Unit (GPU), with CUDA version 9.0. The use of a less advanced GPU was intentional, aiming to demonstrate and develop a model capable of operating on most systems present in volcanic observatories worldwide or on less powerful devices, ensuring efficiency with minimal advanced hardware requirements.

To evaluate the performance of the model, a series of metrics (Kamath et al., 2019) are employed to assess the predictions. Specifically, the accuracy metric is used as the primary criterion to determine whether a new model is an improvement during training on the validation set, with the aim of avoiding overfitting. Additionally, the metrics of precision, recall, and F1-score are also studied to gain further insight into the model's behavior.

The used metrics are commonly derived from the confusion matrix. Based on its elements, these metrics are defined as follows:

1. Accuracy (ACC): Measures the proportion of correctly classified instances (both positive and negative) over the total number of predictions. It provides an overall sense of the model's correctness.

$$\mathrm{ACC} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Number of Events}} \tag{2}$$

2. Precision (PR): Measures the proportion of correctly predicted positive instances out of all instances predicted as positive. It evaluates the quality of positive predictions.

$$\mathrm{PR} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{3}$$

3. Recall (RC): Measures the proportion of correctly predicted positive instances out of all actual positive instances. It evaluates the model's ability to detect positive cases.

$$\mathrm{RC} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{4}$$

4. F1-score (F1): The harmonic mean of precision and recall, which provides a single metric that balances both concerns. It is especially useful when the class distribution is imbalanced.

$$\mathrm{F1} = 2 \cdot \frac{\mathrm{PR} \cdot \mathrm{RC}}{\mathrm{PR} + \mathrm{RC}} \tag{5}$$

4.4. Generalization Tests

To evaluate the adaptability and predictive capability of the model on data not used during training, three additional experiments have been designed.

The first experiment examines data recorded by the same HDAS array during the Cumbre Vieja eruption, but from the month of October. This analysis focuses on continuous monitoring over a 24-hr period to ensure the robustness of the RNN-DAS model. The primary goal is to verify that, despite potential co-eruptive changes in the volcanic system and the resulting seismicity, the detection of events remains accurate. Event detections are compared with the existing seismic catalog derived from traditional seismic stations and the well-known PhaseNet-DAS model (Zhu et al., 2023a). Both models employ a probability threshold of $2/3$ to identify the onset of a phase or event, and a minimum interval of two RNN-DAS windows (9.6 s) between events is enforced. An event is considered valid if at least 500 DAS channels register a phase onset, following the methodology proposed by Zhu et al. (2023a).

The second experiment tests the RNN-DAS model at Stromboli volcano, utilizing a one-hour sample of records capturing explosive events and volcanic tremor (Biagioli et al., 2024) available on Biagioli et al. (2023). Unlike the La Palma study, this DAS system is deployed on land, approximately 1 km away from the volcano, and consists of only 362 usable DAS channels over a 1 km fiber, with a spatial sampling interval of 2.4 m and a gauge length of 5 m. The provided strain rate data has a sampling rate of 200 Hz; therefore, we first applied a

Table 1
Number of Unique Waveforms Used for Model Training, Validation, and Testing[a], Along With per-Frame Test Performance Metrics (in %)

Model	Train	Validation	Test	ACC	PR	RC	F1
RNN-DAS$_{1000}$	2182459	545615	1169174	95.94	93.88	90.63	92.23
RNN-DAS$_{1150}$	+476160	+119040	+148800	97.46	95.13	92.86	93.98

[a]The + indicates the number of additional waveforms used to retrain the first model.

downsampling procedure to match it to 100 Hz, ensuring compatibility with the RNN-DAS model. Explosions at Stromboli are frequent (Ripepe et al., 2008), with energy spanning a broad frequency range (up to 10 Hz) (Neuberg et al., 1994), and are caused by bursts of volcanic gases in the conduit (Blackburn et al., 1976). These explosions release energy both acoustically and as seismic waves, the latter of which are recorded by the DAS array as a specific type of volcanic earthquake. Such earthquakes are typically spindle-shaped and are characterized by emergent onsets without clear S-wave arrivals (Biagioli et al., 2024). The objective of this experiment is to assess the generalization capabilities of the RNN-DAS model when applied to DAS measurements recorded on land and to evaluate its ability to detect seismo-volcanic events with similar energy content but originating from different source mechanisms. This investigation addresses the critical need for developing robust models that can generalize to other volcanic environments and adapt to varying seismic record types with minimal retraining. For this evaluation, the RNN-DAS model predictions were analyzed using a threshold of 0.5 for the predominant class to assess raw model performance. These results were directly compared with PhaseNet-DAS predictions. Considering that an event is detected following the same approach as the first experiment, but proportionally reducing the required number of simultaneously identified channels with events to 50 for the new DAS data.

The final experiment is designed with two primary objectives. First, to evaluate the temporal behavior of the model in terms of prediction latency, and second, to investigate the minimum detection thresholds in terms of SNR, event magnitude, and hypocentral distance. To this end, we selected 500 events from the three months of coeruptive data available for the La Palma eruption. For each event, a 3-min sample was extracted and centered on the origin time reported in the seismic catalog. For the first objective, we assess whether the model is capable of generating predictions in less time than the duration of the input signal, a key requirement for real-time implementation. This evaluation was performed globally and at the per-channel level, breaking down the total inference time into the following steps: data reading, preprocessing, feature extraction, prediction, and overall processing time. The system is considered to meet real-time constraints if, for all cases, the total processing time is shorter than the input duration with 95% confidence ($\alpha = 0.05$) based on the empirical distribution. For the second objective, we analyzed the model's ability to detect events across a range of SNRs, magnitudes, and hypocentral distances. We imposed a detection criterion requiring that at least 50 DAS channels correctly register the event. This allowed us to explore the performance limits of the model when detecting low-magnitude, distant, or strongly attenuated events that are primarily observable only in the DAS channels closest to the volcano. This is especially relevant given that the DAS array is deployed approximately 10 km away from the volcanic edifice. We also examined the distribution of predicted probabilities for each event using a threshold of $2/3$ and the same grammar-based function as in the first generalization experiment.

5. Experimental Results

5.1. Performance of the Model

First, the total number of waveforms used and the results of the metrics obtained for both trained models are shown in Table 1. These results refer to per-frame predictions and demonstrate good performance, with values exceeding 90% in all cases. For both models, the RC and PR metrics exhibit lower performance, which can be attributed to the frames corresponding to the coda of the events. The coda corresponds to the attenuated portion of the seismic signal, generated by scattering through the medium during its propagation, making it more similar to background seismic noise. This leads, in the case of RC, to errors caused by the insertion of false negatives when classifying that portion as noise. Similarly, the lower performance in PR can be explained by the introduction of false positives in the coda portion, where frames of seismic noise are confused with coda frames. This hypothesis

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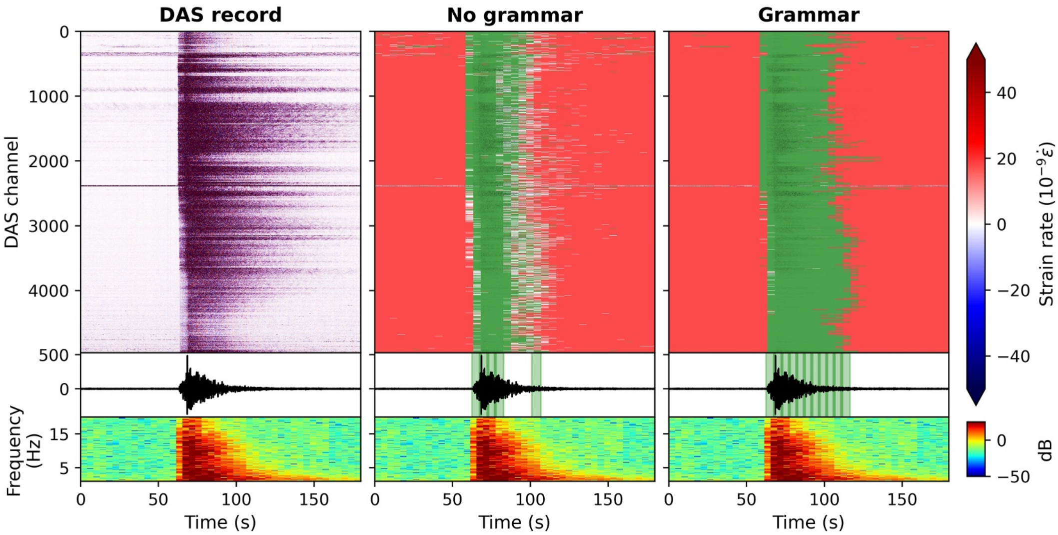

Figure 3. Example of the RNN-DAS model prediction with and without the grammar function applied. The original recording is 3 min long and corresponds to an event with $M_l = 4.22$. In the prediction outputs, the windows identified in green correspond to the VT class, those in red to noise type 1, and those in gray to noise type 2. If the 2/3 probability threshold for any class is not exceeded within a given window, the window is displayed as transparent. The normalized signal is shown alongside the straingram and spectrogram of channel 3500, with the windows identified as VT for that channel highlighted in green.

is supported by the improved performance of all metrics in the RNN-DAS (1150) model, which, after being retrained using only seismic noise samples, is able to detect and differentiate coda frames from background noise with higher accuracy. Despite these potential frame-level errors, the metrics' performance is high in both cases, demonstrating the ability of the RNN-LSTM architecture to model time series with spatial coherence recorded by the HDAS. The RNN-DAS (1150) model was selected as the better-performing model for the rest of the study.

The model outputs, for a DAS recording, a series of class-normalized probabilities for each channel and time frame. To further leverage the capabilities offered by the DAS technique for detection and to reduce potential prediction errors across channels or with the event's coda, a grammar function was developed to act on these class output probabilities (see Figure 3). This function operates in two distinct ways.

First, the method ensures consistency across channels by applying a filter to the probabilities of the VT or low-energy seismic noise classes for each time frame across all channels. If a threshold number of frames within a group of channels predict the same class, all frames that either predict a different class or have a predicted probability lower than a predefined threshold will be reassigned to the majority class within that group. This threshold value is manually set in advance to guide the reassignment process. Thereby ensuring robustness against individual frame prediction errors. Note that the second class of noise is not considered in this functionality, as it is specifically used to identify channels affected by high-amplitude noise components resulting from site effects or anthropogenic sources. The second functionality is designed to improve coda prediction based on the previously discussed issues. Using a trigger-on and trigger-off approach with VT class probabilities across frames of a channel, probabilities are adjusted for frames that, after detecting a VT event with a channel-specific probability above the trigger-on threshold, are assigned to VT classes, even if the predominant class is seismic noise, as long as the VT class prediction probability is above the trigger-off threshold. This mechanism enhances the detection of the seismic coda. This grammar function has proven to be a useful tool for improving the raw predictions of the model. For the La Palma case, it was found that frame probability thresholds of 2/3 for 50% of the channels within groups of 10 channels (100 m of monitoring), with trigger-on and trigger-off thresholds for the coda of 0.9 and 0.05, respectively, provide satisfactory results.

Figure 4 illustrates examples of the model's performance on different types of VT event recordings from the month of November. The model demonstrates excellent detection capability in all cases and across the majority of DAS channels, with the use of the grammar function enhancing both coda detection and coherence.

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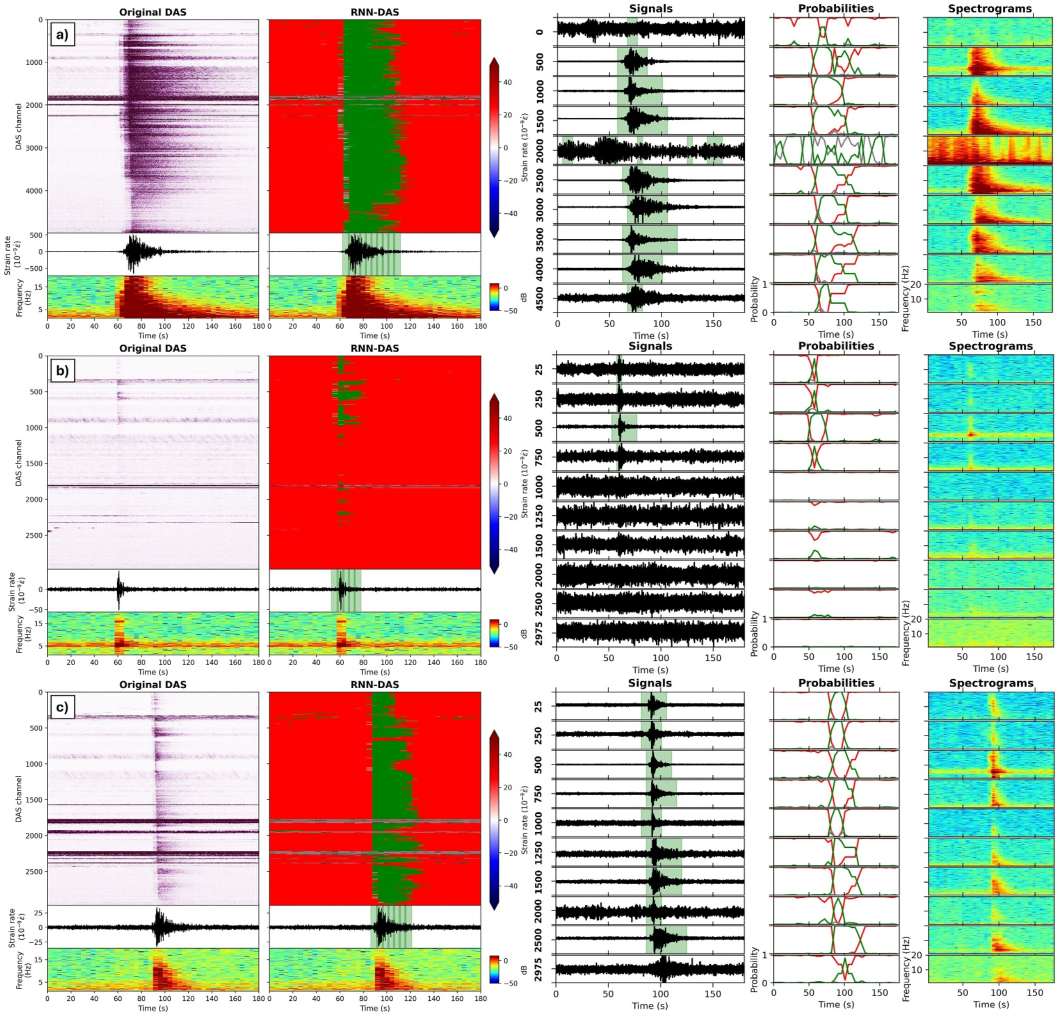

Figure 4. Examples of RNN-DAS model performance on different VT event recordings. Left: The original recording with a green dashed line marking the event origin time from the seismic catalog, alongside the prediction outputs for the three classes (green = VT, red = low-energy noise, gray = high-energy noise), showcasing the straingram from channel 1500 and 500 in (b). Right: Straingrams from various DAS channels with the regions identified as the VT class highlighted in green, the temporal evolution of the output probabilities per class provided by the model, and the spectrogram for each channel. (a) Attenuated event with magnitude $M_l = 3.76$. (b) Event recorded with low SNR and magnitude $M_l = 2.28$. (c) Nearby event recorded with high SNR and magnitude $M_l = 2.75$.

It is worth noting that, despite the presence of high-amplitude seismic noise that can mask the signal, the model remains capable of accurately distinguishing the event. This is because the model predicts using the estimated energy based on frequency, allowing it to differentiate the event from the noise, even though its identification becomes more complex, as evidenced in case b) where the event is highly attenuated and exhibits significant energy in only a handful of DAS channels.

Another case of interest when evaluating the model's performance is a sequence of events consisting of a main shock followed by lower-magnitude aftershocks. Figure 5 presents such a sequence, where a primary event is followed by a large number of smaller events within a short time interval.

It is observed that, despite the events occurring in close temporal proximity and exhibiting low SNR, the vast majority are correctly identified across most DAS channels. Only some events, which are significantly attenuated and detectable only by stations near the volcano, are not accurately detected by our model in the DAS records. Nevertheless, the results demonstrate a strong performance of the RNN-DAS model.

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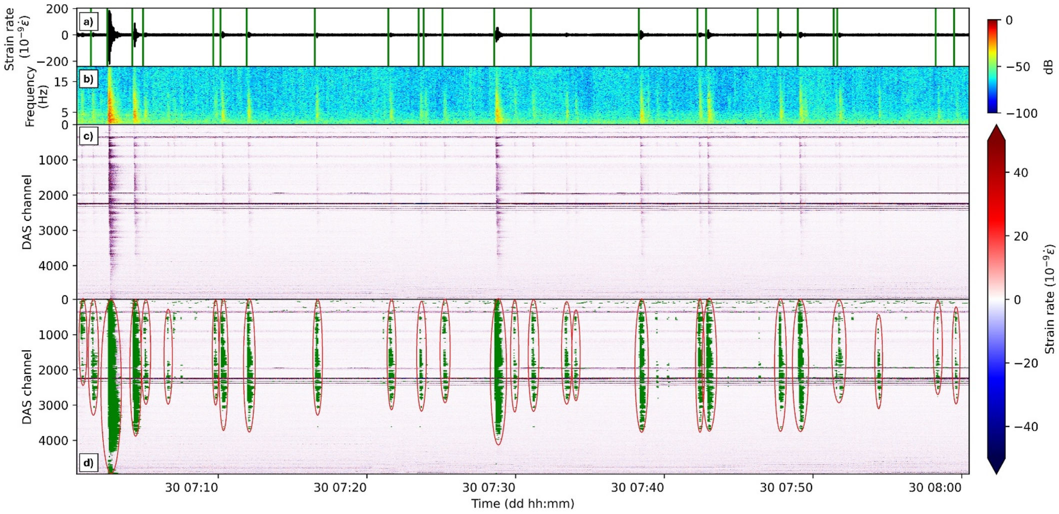

Figure 5. Example of the model's performance on a one-hour sample from November 30, between 07:00 and 08:00, featuring a main event of $M_l = 3.22$. (a) Straingram of channel 1500 with green lines indicating the origin time detected in the seismic catalog, along with its spectrogram in (b). (c) DAS raw signal and in (d) the DAS signal with frames predicted as events shown in green, using a probability threshold of 2/3, and events detected in at least 500 DAS channels highlighted with a red oval.

5.2. Generalization Capabilities

First, a generalization experiment of the model was conducted using data from October 29, during the eruption, as this day provides a complete and uninterrupted datastream. The results of applying the RNN-DAS and PhaseNet-DAS models for continuous monitoring over a one-hour period on this day are shown in Figure 6. Both models used the same probability threshold to identify phases or events in each channel.

The results indicate that the RNN-DAS model successfully detects the majority of events listed in the seismic catalog, with the exception of highly attenuated signals and regional events that are not included in the catalog. The PhaseNet-DAS model also performs well, accurately identifying events that exhibit clear P and S phases. However, it shows a slightly higher tendency to produce false positives, particularly in the prediction of S phases.

Close-up examples comparing both models are shown in Figures 7a–7d. In general, both models demonstrate comparable performance. Figure 7e shows the number of detected events for each model compared to the seismic catalog. The primary differences are observed in a slight overestimation of S-phase picks by PhaseNet-DAS and an underestimation of the number of detected channels for highly attenuated events without significant energy by RNN-DAS. When quantified using the Jaccard similarity index (Fletcher & Islam, 2018), both models show agreement with the catalog, with RNN-DAS achieving better performance compared to PhaseNet-DAS detections based on P phases, S phases, and their combined detection rates.

This behavior may be attributed to differences in ambient noise characteristics and conditions of the submarine DAS array deployed at La Palma. In such environments, attenuation and complex wave propagation effects typical of volcanic settings are likely to be more pronounced than in the inland volcanic environments where PhaseNet-DAS was originally trained. Consequently, the CNN-based architecture of PhaseNet-DAS may face challenges generalizing effectively under these different SNR conditions, especially when identifying S phases. In contrast, the RNN-based architecture of RNN-DAS, which leverages engineered energy-based features, appears more robust at distinguishing seismic events from background noise in this setting.

These results showcase that the RNN-DAS model, despite being applied to different time periods than those used for training, where the existing seismicity may differ due to the evolution of the volcanic system, still performs well. This highlights its potential as a volcano monitoring tool over extended periods, during which both the volcanic system and seismicity may evolve, making the model a valuable tool for forecasting.

For the second generalization experiment, no grammar function was employed in order to assess the raw capabilities of the model in this new volcanic setting (Figure 8a). The prediction results are shown in Figure 8. It was found that RNN-DAS is capable of satisfactorily detecting the presence of events with high SNR (Figures 8b and

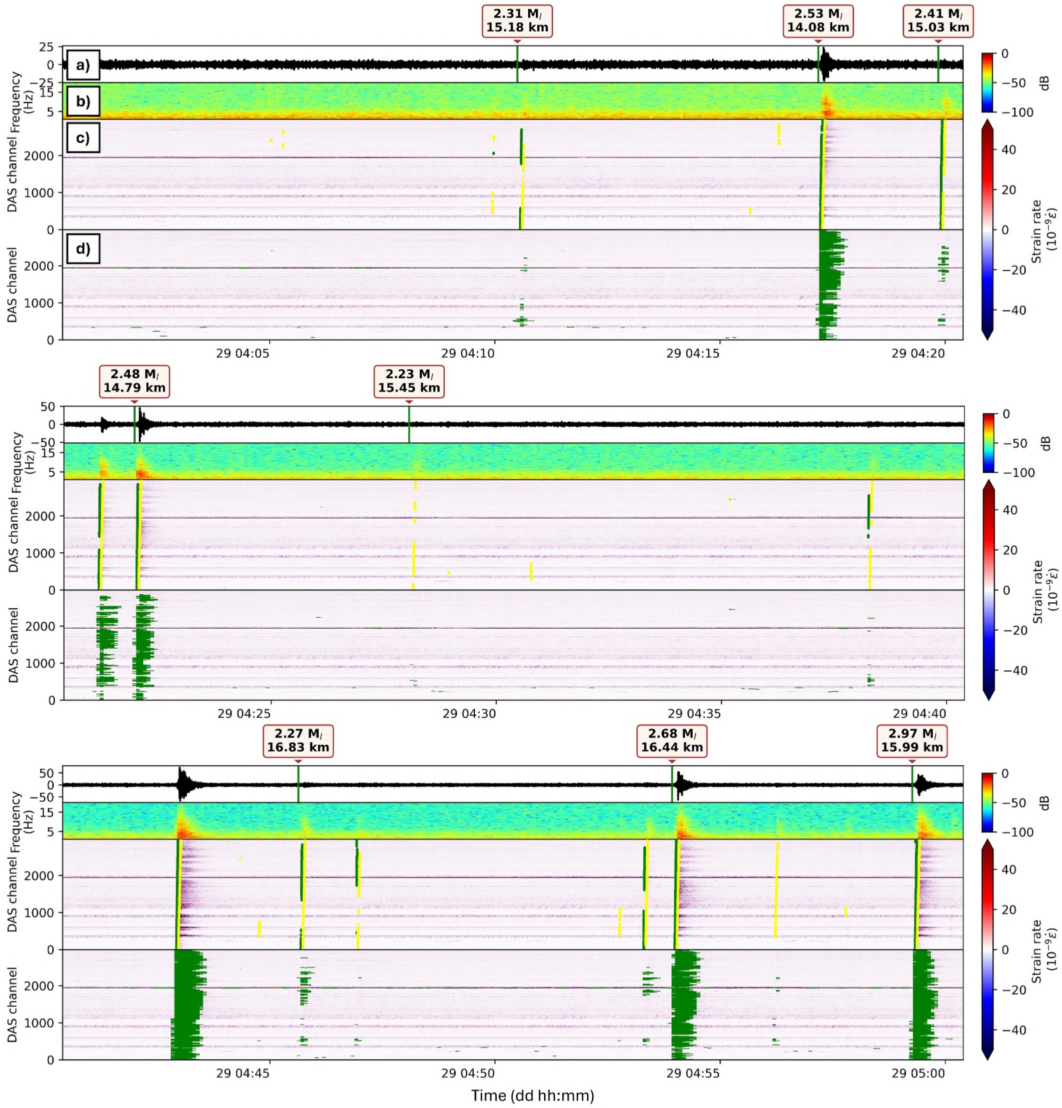

Figure 6. Prediction results for continuous monitoring during the 04:00–05:00 hr on October 29. (a) Straingram from channel 1500 with catalog events marked by their origin time, magnitude, and hypocentral distance to the DAS array. (b) Spectrogram from channel 1500. (c) Prediction results from the PhaseNet-DAS model, with P phases in green and S phases in yellow. (d) Results from the RNN-DAS model, with event duration windows in green.

8d) as well as their absence in records containing only volcanic tremor. In the case of smaller events with low SNR (Figure 8c), the model still identifies the events; however, due to the high-amplitude volcanic tremor, the events are masked, which hinders the model's ability to detect them across all channels (Figure 8c). Nonetheless, the results for detecting explosive events (Figures 8e and 8f) are similar to those reported by Biagioli et al. (2024), with the appearance of scattered false positives across some channels due to raw low-probability predictions, which arise specifically from not applying the model's grammar function. For instance, the PhaseNet-DAS model does not accurately recognize the presence of these events (Figure 8g), due to the clear absence of seismic phases and the significant high-amplitude tremor that masks the waveforms.

It is important to note that, while RNN-DAS can detect seismic events associated with volcanic explosions, it cannot distinguish between seismic noise and volcanic tremor, nor can it differentiate volcanic explosions from purely tectonic (VT) events. This limitation arises because the model has not been trained to differentiate between these types of events. However, this issue could easily be addressed by retraining the model and redefining the prediction classes to better adapt to the characteristic events of each volcano.

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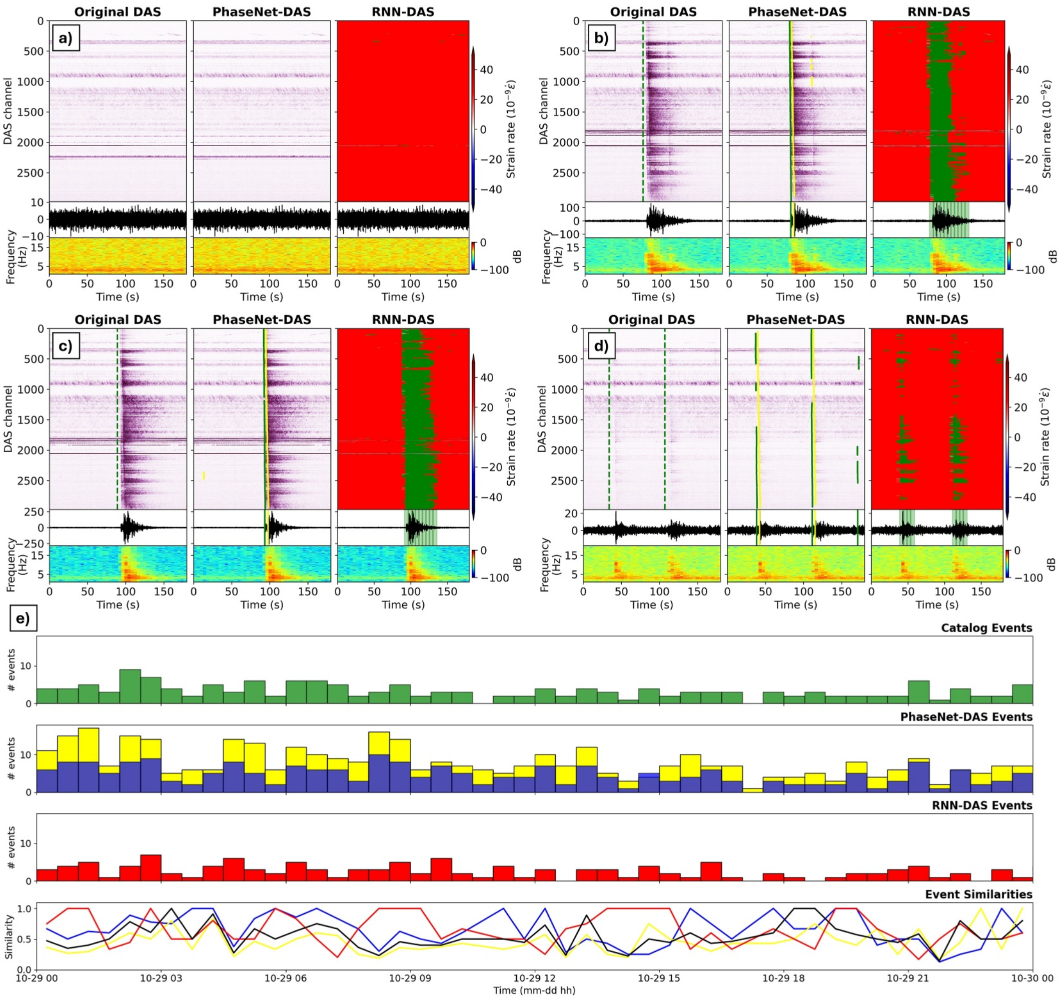

Figure 7. Comparison of PhaseNet-DAS and RNN-DAS model predictions over the 24-hr tested period. (a) Sample containing only seismic noise. (b) Event with magnitude $M_l = 3.22$. (c) Event with magnitude $M_l = 3.17$. (d) Events with magnitude $M_l = 2.42$ and $M_l = 2.49$. (e) Bar plot of catalog events (green), events identified by PhaseNet-DAS (P in blue, S in yellow, and black for joint prediction), and RNN-DAS (red) in 1-hr intervals, along with similarity indices with the catalog.

For the third generalization test of the model, 500 events were randomly selected from the three months of coeruptive data, and the results are shown in Figure 9.

First, when comparing the results from both generalization tests for RNN-DAS predictions in La Palma and Stromboli (see Figure 9a), it is observed that there are differences in the distributions of event durations and slight differences in the distributions of detection probabilities. These differences can be attributed to the distinct source mechanisms of the events despite their similarities in energy or frequency range, with VT events from La Palma having a higher detection probability compared to volcanic explosions from Stromboli. Thus suggesting that the model's ability to distinguish between these events could be further enhanced through retraining, enabling it to better differentiate between various volcano-seismic event types. This refinement would improve the model's performance and adaptability, allowing it to more accurately identify and classify different events detected by DAS systems in active volcanic regions.

Second, the study on the detection limitations of the RNN-DAS model is presented in Figure 9b. It is observed that most events are successfully detected, while undetected events are those with a SNR approximately below 1.5 dB with respect to the background noise preceding the event. Which means that the model is capable of detecting events whose signal energy is at least 1.41 times greater than the preceding seismic noise. In particular, events with magnitude greater than 2 are successfully detected in most cases. Lower-magnitude events, which are

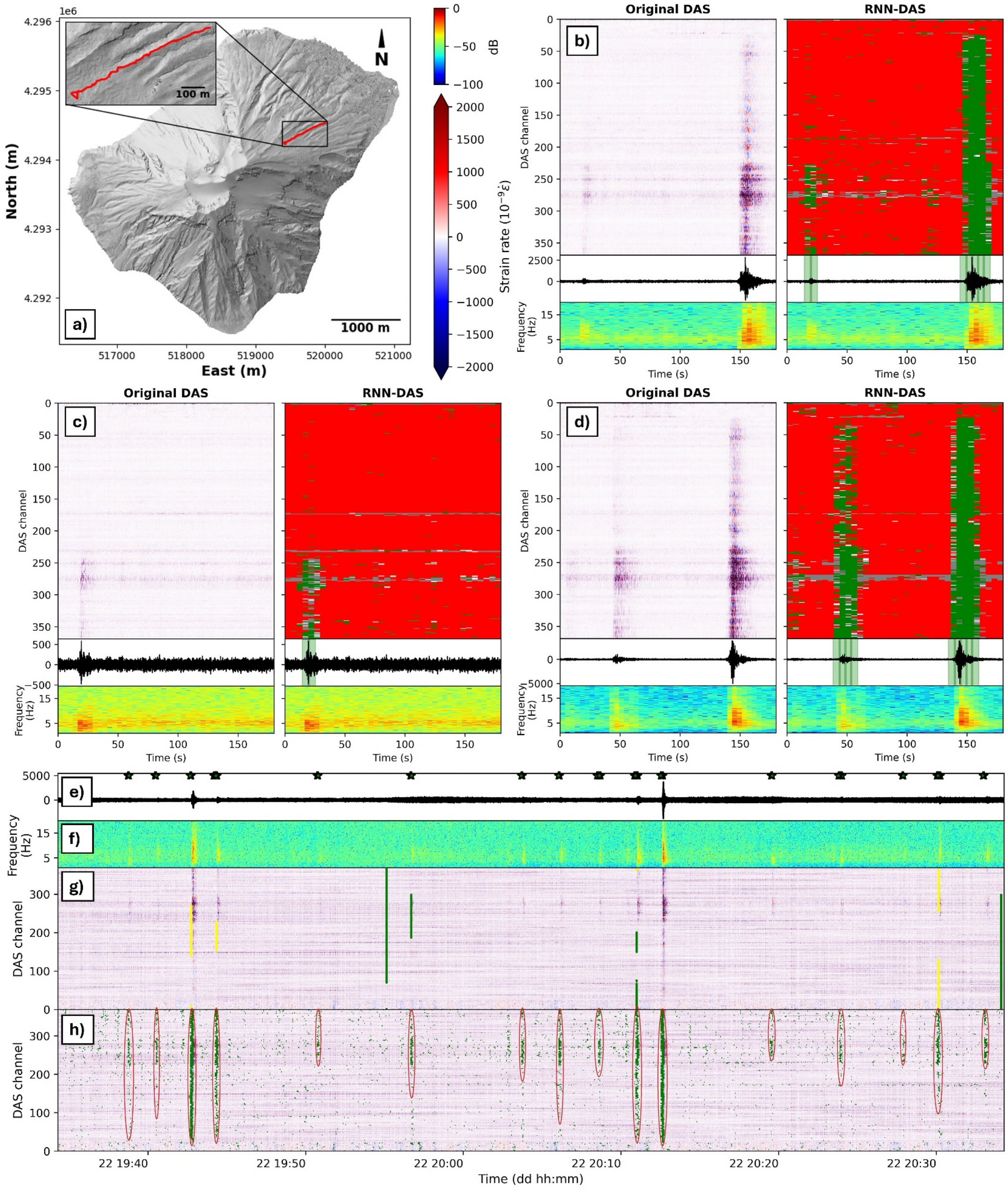

Figure 8. Performance of the RNN-DAS model on DAS data recorded at Stromboli. (a) Map of Stromboli island (Bisson et al., 2025) with the DAS array in red (Biagioli et al., 2024). (b, d) Illustrate the prediction for events with high SNR. (c) Shows the prediction for an event that is masked by volcanic tremor. (g, h) Present the continuous predictions over the entire DAS record for the PhaseNet-DAS (with P-phases in green and S-phases in yellow) and RNN-DAS (highlighted in red ovals where an event is detected), respectively. The displayed channel in (e) corresponds to channel 350, with its spectrogram shown in (f), where the event predictions for this channel are highlighted in green, and a green star marks each window in which RNN-DAS detects an event across all DAS channels.

inherently less energetic, are also detected provided they are not significantly attenuated due to their hypocentral distance from the DAS array. The distribution of events with respect to distance is relatively homogeneous, indicating that the model is capable of detecting distant events as long as their energy (and consequently their SNR) is sufficiently high.

These results demonstrate the robustness of RNN-DAS, which is capable of correctly detecting events with signal energy only 1.5 dB above the background noise in at least 50 DAS channels. For other DAS arrays, such as the

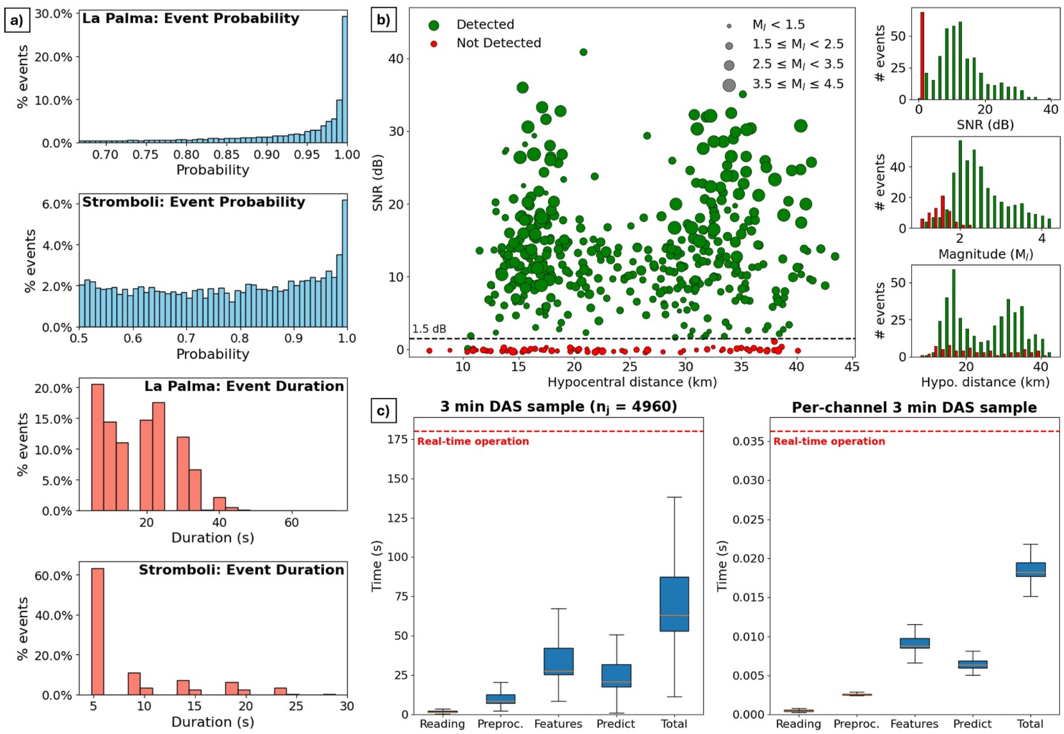

Figure 9. (a) Bar plots showing the distribution of mean detection probability and detection duration per DAS channel for La Palma (top) and Stromboli (bottom), obtained from the RNN-DAS model generalization tests. The *y*-axis indicates the percentage of detections corresponding to each interval. (b) Main panel: results from the third generalization experiment, displaying the average SNR in dB versus hypocentral distance to the DAS array in La Palma. Each point represents an event, with color indicating detection status (green: detected, red: not detected) and marker size proportional to event magnitude. The black line denotes the approximate the minimum SNR threshold found separating detected event signals from background seismic noise. On the right, bar charts illustrate the distribution of detected (green) and not detected (red) events as a function of SNR, magnitude, and hypocentral distance. (c) Temporal performance analysis of the RNN-DAS model under real-time constraints based on the third experiment. On the left, total inference time per event is shown when processing three 3 min samples (up to 4960 channels). On the right, the corresponding time taken per individual DAS channel is presented. In both cases, the total time is broken down into reading, preprocessing, feature extraction, and prediction stages. Boxplots indicate the interquartile range (blue), median (orange), and whiskers encompassing 95% of the distribution.

one deployed in Stromboli, which is closer to the event sources, the expected attenuation would be lower, resulting in higher signal energy and, consequently, improved detection capabilities. Nevertheless, the RNN-DAS model has proven to perform robustly even under challenging conditions, with the main limitation of the model being the detection of highly attenuated events whose signal energy barely rises above the noise level by the time they reach the DAS array.

Third, the inference time analysis is presented in Figure 9c. It is observed that even when using all available DAS channels, up to 4960 in the cases with the longest interrogated fiber sections, the distribution of total inference times remains below the 180 s threshold, which corresponds to the duration of the input signal. Moreover, all evaluated cases remain under the real-time operation limit within the 95% confidence interval, as indicated by the boxplots. As expected, the most time-consuming step is the feature extraction stage, where signal features are computed for model input. Nonetheless, the overall inference time remains far below the real-time operation limit. Since the global inference time is naturally dependent on the number of DAS channels, we also analyzed the per-channel computation times. The results are similarly favorable, with total per-channel processing times well below the real-time constraint.

The obtained results demonstrate that our model is capable of operating under real-time constraints for large-scale DAS systems with thousands of channels. While inference time will naturally vary depending on the computing infrastructure, all tests were performed on a machine with components from 2017. Importantly, the prediction process does not require GPU acceleration, making it suitable for deployment on most modern systems across volcanological observatories without the need for high-performance computing resources.

6. Conclusions

In this paper, we have demonstrated how RNNs with LSTM can be applied to the study of seismovolcanic signals for volcanic monitoring purposes using the DAS technique. To achieve this, we developed a new deep learning model, RNN-DAS, designed to operate in real-time for detecting volcanic-tectonic events and recording their waveforms. The model leverages features extracted from the parametrization of these signals into energy coefficients, which account for both the temporal evolution and spatial coherence provided by ultra-dense DAS measurements.

The results show that the model achieves over 90% in all studied metrics, yielding performance comparable to visual inspection and surpassing other existing models. Trained with a large data set of VT events from an underwater high-fidelity DAS during the La Palma 2021 eruption, the model has proven capable of detecting volcanic events in generalization tests across different time periods. It has demonstrated reliability for the same volcano, as well as adaptability for others, such as Stromboli. Despite changes in source mechanisms, medium effects, or in the case of land-based DAS, the model continues to provide accurate predictions on volcano-seismic events recorded by DAS arrays. The only limiting factor is the signal energy, with the model capable of detecting events whose energy is just slightly above the background seismic noise.

This advancement addresses key challenges in volcanic seismology, particularly in managing the vast data sets produced by DAS, and represents a significant step forward by enabling automated, scalable, and real-time analysis of volcanic signals. Looking ahead, several promising lines of research are proposed to further enhance the model's adaptability. First, testing its real-time applicability with other DAS data sets of VT events from active volcanoes will broaden its monitoring capabilities. Second, retraining the model with new DAS data from different types of volcano-seismic events, such as those observed at Stromboli, will ensure more comprehensive and accurate classification in diverse volcanic environments by redefining classification classes to better accommodate the different types of events detected. In addition, the use of a larger number of DAS channels simultaneously for energy-based feature extraction could be explored to further improve detection performance. Furthermore, the development of an automatic method that adds the localization and magnitude estimation of detected events represents a promising future research direction that is already underway.

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Data Availability Statement

The data sets used in this study are:

1. HDAS data from La Palma—Data sets containing the HDAS data corresponding to Figures 3–7 (for panels a), b), c) and d)) are available at Fernandez-Carabantes et al. (2025b).
2. DAS data from Stromboli—This data set is publicly available in Biagioli et al. (2023).

The digital elevation model of La Palma used in Figure 1 is available from Copernicus (2022), while the one used in Figure 8 for Stromboli from Bisson et al. (2025). All data processing and figure generation were conducted using Python version 3.7 using the Matplotlib environment (Caswell et al., 2020).

The PhaseNet-DAS model is available at Zhu et al. (2023b).

The RNN-DAS model is available at Fernandez-Carabantes et al. (2025a) (https://doi.org/10.5281/zenodo.15858493).

References

Afnimar, Hidayat, A., Kristianto, Triastuty, H., Basuki, A., & Indrastuti, N. (2022). Source mechanism of volcanic tectonic events from October 2010 to December 2011 at the Sinabung volcano area. *Geoscience Letters*, *9*(1), 28. https://doi.org/10.1186/s40562-022-00235-w

Allen, R. V. (1978). Automatic earthquake recognition and timing from single traces. *Bulletin of the Seismological Society of America*, *68*(5), 1521–1532. https://doi.org/10.1785/bssa0680051521

Almendros, J., Chouet, B., & Dawson, P. (2001). Spatial extent of a hydrothermal system at Kilauea volcano, Hawaii, determined from array analyses of shallow long-period seismicity: 1. Method. *Journal of Geophysical Research*, *106*(B7), 13565–13580. https://doi.org/10.1029/2001jb000310

Almendros, J., Chouet, B., Dawson, P., & Bond, T. (2002). Identifying elements of the plumbing system beneath Kilauea volcano, Hawaii, from the source locations of very-long-period signals. *Geophysical Journal International*, *148*(2), 303–312. https://doi.org/10.1046/j.1365-246x.2002.01010.x

Almendros, J., Ibáñez, J. M., Carmona, E., & Zandomeneghi, D. (2007). Array analyses of volcanic earthquakes and tremor recorded at Las Cañadas caldera (Tenerife Island, Spain) during the 2004 seismic activation of Teide volcano. *Journal of Volcanology and Geothermal Research*, *160*(3–4), 285–299.

Álvarez, I., García, L., Mota, S., Cortés, G., Benítez, C., & De la Torre, Á. (2013). An automatic p-phase picking algorithm based on adaptive multiband processing. *IEEE Geoscience and Remote Sensing Letters*, *10*(6), 1488–1492. https://doi.org/10.1109/lgrs.2013.2260720

Baba, S., Araki, E., Yokobiki, T., Kawamata, K., Uchiyama, K., & Yoshizuka, T. (2024). Seismic observation using distributed acoustic sensing around the Tsugaru Strait at the Japan and Kuril trenches, northeastern Japan. *Earth Planets and Space*, *76*(28), 28. https://doi.org/10.1186/s40623-024-01975-z

Balluff, S., Bendfeld, J., & Krauter, S. (2020). Meteorological data forecast using RNN. In *Deep Learning and Neural Networks: Concepts, Methodologies, Tools, and Applications* (pp. 905–920). IGI Global.

Barrancos, J., D'Auria, L., Padilla, G., Preciado-Garbayo, J., & Pérez, N. M. (2022). Hdas (high-fidelity distributed acoustic sensing) as a monitoring tool during 2021 Cumbre Vieja eruption. In *EGU General Assembly Conference Abstracts* (pp. EGU22–5327).

Benítez, C., Ramirez, J., Segura, J. C., Rubio, A., Ibanez, J. M., Almendros, J., & García-Yeguas, A. (2006). Continuous hmm-based volcano monitoring at deception island, Antarctica. In *2006 IEEE International Conference on Acoustics Speech and Signal Processing Proceedings* (Vol. 5). IEEE.

Berz, G., Kron, W., Loster, T., Rauch, E., Schimetschek, J., Schmieder, J., et al. (2001). World map of natural hazards–a global view of the distribution and intensity of significant exposures. *Natural Hazards*, *23*(2–3), 443–465. https://doi.org/10.1023/a:1011193724026

Beyreuther, M., & Wassermann, J. (2008). Continuous earthquake detection and classification using discrete hidden Markov models. *Geophysical Journal International*, *175*(3), 1055–1066. https://doi.org/10.1111/j.1365-246X.2008.03938.x

Biagioli, F., Métaxian, J.-P., Stutzmann, E., Ripepe, M., Bernard, P., Trabattoni, A., et al. (2023). Replication data for: Array analysis of seismo-volcanic activity with distributed acoustic sensing [Dataset]. *IPGP Research Collection*. https://doi.org/10.18715/IPGP.2023.lgl6bxby

Biagioli, F., Métaxian, J.-P., Stutzmann, E., Ripepe, M., Bernard, P., Trabattoni, A., et al. (2024). Array analysis of seismo-volcanic activity with distributed acoustic sensing. *Geophysical Journal International*, *236*(1), 607–620. https://doi.org/10.1093/gji/ggad427

Biondi, E., Li, J., Saunders, J., Husker, A., & Zhan, Z. (2024). The frontiers of distributed acoustic sensing for seismological applications. In *Copernicus meetings*.

Biondi, E., Zhu, W., Li, J., Williams, E. F., & Zhan, Z. (2023). An upper-crust lid over the long valley magma chamber. *Science Advances*, *9*(42), eadi9878. https://doi.org/10.1126/sciadv.adi9878

Bisson, M., Gianardi, R., Civico, R., Madonia, P., Ricci, T., & Spinetti, C. (2025). Lidar-derived digital surface model of Stromboli Island updated to August 2023. *Scientific Data*, *12*(1), 522. https://doi.org/10.1038/s41597-025-04856-6

Blackburn, E. A., Wilson, L., & Sparks, R. S. J. (1976). Mechanisms and dynamics of strombolian activity. *Journal of the Geological Society*, *132*(4), 429–440. https://doi.org/10.1144/gsjgs.132.4.0429

Bormann, P., & Wielandt, E. (2013). Seismic signals and noise. In *New Manual of Seismological Observatory Practice 2 (NMSOP2)* (pp. 1–62). Deutsches GeoForschungsZentrum GFZ.

Brown, S. K., Jenkins, S. F., Sparks, R. S. J., Odbert, H., & Auker, M. R. (2017). Volcanic fatalities database: Analysis of volcanic threat with distance and victim classification. *Journal of Applied Volcanology*, *6*, 1–20. https://doi.org/10.1186/s13617-017-0067-4

Canario, J. P., Mello, R., Curilem, M., Huenupan, F., & Rios, R. (2020). In-depth comparison of deep artificial neural network architectures on seismic events classification. *Journal of Volcanology and Geothermal Research*, *401*, 106881. https://doi.org/10.1016/j.jvolgeores.2020.106881

Carracedo, J. C., Troll, V. R., Day, J. M., Geiger, H., Aulinas, M., Soler, V., et al. (2022). The 2021 eruption of the Cumbre Vieja volcanic ridge on La Palma, Canary Islands. *Geology Today*, *38*(3), 94–107. https://doi.org/10.1111/gto.12388

Caswell, T. A., Droettboom, M., Lee, A., Hunter, J., Firing, E., Stansby, D., et al. (2020). Matplotlib v3.2.1 [Software]. *Zenodo*. https://doi.org/10.5281/zenodo.3714460

Choi, E., Schuetz, A., Stewart, W. F., & Sun, J. (2017). Using recurrent neural network models for early detection of heart failure onset. *Journal of the American Medical Informatics Association*, *24*(2), 361–370. https://doi.org/10.1093/jamia/ocw112

Chouet, B. (2003). Volcano seismology. *Pure and Applied Geophysics*, *160*(3), 739–788. https://doi.org/10.1007/PL00012556

Chouet, B., Saccorotti, G., Martini, M., Dawson, P., De Luca, G., Milana, G., & Scarpa, R. (1997). Source and path effects in the wave fields of tremor and explosions at Stromboli volcano, Italy. *Journal of Geophysical Research*, *102*(B7), 15129–15150. https://doi.org/10.1029/97JB00953

Chouet, B. A., & Matoza, R. S. (2013). A multi-decadal view of seismic methods for detecting precursors of magma movement and eruption. *Journal of Volcanology and Geothermal Research*, *252*, 108–175. https://doi.org/10.1016/j.jvolgeores.2012.11.013

Copernicus Emergency Management Service—Mapping. (2022). Emsr546. Retrieved from https://emergency.copernicus.eu/mapping/list-of-components/EMSR546

Corsaro, M., Cannavò, F., Currenti, G., Palazzo, S., Allegra, M., Jousset, P., & Spampinato, C. (2024). Seismic phases picking with artificial intelligence: A novel approach for distributed acoustic sensing data analysis. In *Copernicus Meetings*.

Currenti, G., Allegra, M., Cannavò, F., Jousset, P., Prestifilippo, M., Napoli, R., et al. (2023). Distributed dynamic strain sensing of very long period and long period events on telecom fiber-optic cables at Vulcano, Italy. *Scientific Reports*, *13*(1), 4641. https://doi.org/10.1038/s41598-023-31779-2

Currenti, G., Jousset, P., Napoli, R., Krawczyk, C., & Weber, M. (2021). On the comparison of strain measurements from fibre optics with dense seismometer array at Etna volcano (Italy). *Solid Earth Discussions*, 1–19.

D'Auria, L., Koulakov, I., Prudencio, J., Cabrera-Pérez, I., Ibáñez, J. M., Barrancos, J., et al. (2022). Rapid magma ascent beneath La Palma revealed by seismic tomography. *Scientific Reports*, *12*(1), 17654. https://doi.org/10.1038/s41598-022-21818-9

De Barros, L., Bean, C. J., Zecevic, M., Brenguier, F., & Peltier, A. (2013). Eruptive fracture location forecasts from high-frequency events on Piton de la Fournaise volcano. *Geophysical Research Letters*, *40*(17), 4599–4603. https://doi.org/10.1002/grl.50890

Del Fresno, C., Cesca, S., Klügel, A., Domínguez Cerdeña, I., Díaz-Suárez, E. A., Dahm, T., et al. (2023). Magmatic plumbing and dynamic evolution of the 2021 La Palma eruption. *Nature Communications*, *14*(1), 358. https://doi.org/10.1038/s41467-023-35953-y

Dondurur, D. (2018). *Acquisition and processing of marine seismic data*. Elsevier.

Dreier, R., Widmer, R., Schick, R., & Wielandt, E. (1994). Stacking of broadband seismograms at Stromboli. *Acta Vulcanologica*, *5*, 165–172.

Durek, J. J., & Ekström, G. (1996). A radial model of anelasticity consistent with long-period surface-wave attenuation. *Bulletin of the Seismological Society of America*, *86*(1A), 144–158. https://doi.org/10.1785/bssa08601a0144

Esposito, A. M., Giudicepietro, F., Scarpetta, S., & Khilnani, S. (2018). A neural approach for hybrid events discrimination at Stromboli volcano. In *Multidisciplinary Approaches to Neural Computing* (pp. 11–21). Springer. https://doi.org/10.1007/978-3-319-96079-0_2

Fakhruzi, I., Titos, M., Benítez, C., & García, L. (2025). Urban traffic monitoring through distributed acoustic sensing: Trial analysis of a potent monitoring tool. *Measurement*, *253*(Part D), 117668. https://doi.org/10.1016/j.measurement.2025.117668

Fernandez-Carabantes, J., Titos, M., D'Auria, L., García, J., García, L., & Benítez, C. (2025a). Javier-fernandezcarabantes/rnn-das: Rnn-das v1.0.0 (v1.0.0) [Software]. *Zenodo*. https://doi.org/10.5281/zenodo.15858493

Fernandez-Carabantes, J., Titos, M., D'Auria, L., García, J., García, L., & Benítez, C. (2025b). RNN-DAS: A new deep learning approach for detection and real-time monitoring of volcano-tectonic events using distributed acoustic sensing [Dataset]. *Zenodo*. https://doi.org/10.5281/zenodo.15105596

Fichtner, A., Hofstede, C., Kennett, B. L., Nymand, N. F., Lauritzen, M. L., Zigone, D., & Eisen, O. (2023). Fiber-optic airplane seismology on the northeast Greenland ice stream. *The Seismic Record*, *3*(2), 125–133. https://doi.org/10.1785/0320230004

Fletcher, S., & Islam, M. Z. (2018). Comparing sets of patterns with the Jaccard index. *Australasian Journal of Information Systems*, *22*. https://doi.org/10.3127/ajis.v22i0.1538

Graves, A., Mohamed, A. R., & Hinton, G. (2013). Speech recognition with deep recurrent neural networks. In *2013 IEEE International Conference on Acoustics, Speech and Signal Processing* (pp. 6645–6649). IEEE.

Gregor, K., Danihelka, I., Graves, A., Rezende, D., & Wierstra, D. (2015). Draw: A recurrent neural network for image generation. In *International Conference on Machine Learning* (pp. 1462–1471). PMLR.

Hartog, A. H. (2017). *An introduction to distributed optical fibre sensors*. CRC Press.

Hewitt, K. (2014). *Regions of risk: A geographical introduction to disasters*. Routledge.

Hinton, G., Deng, L., Yu, D., Dahl, G. E., Mohamed, A. R., Jaitly, N., et al. (2012). Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups. *IEEE Signal Processing Magazine*, *29*(6), 82–97. https://doi.org/10.1109/msp.2012.2205597

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, *9*(8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

Ibáñez, J. M., Benítez, C., Gutiérrez, L. A., Cortés, G., García-Yeguas, A., & Alguacil, G. (2009). The classification of seismo-volcanic signals using hidden Markov models as applied to the Stromboli and Etna volcanoes. *Journal of Volcanology and Geothermal Research*, *187*(3–4), 218–226. https://doi.org/10.1016/j.jvolgeores.2009.07.004

Ibáñez, J. M., Morales, J., Alguacil, G., Almendros, J., & Del Pezzo, E. (1997). Intermediate-focus earthquakes under South Shetland Islands (Antarctica). *Geophysical Research Letters*, *24*(5), 531–534. https://doi.org/10.1029/97gl00314

Ibáñez, J. M., Pezzo, E. D., Almendros, J., La Rocca, M., Alguacil, G., Ortiz, R., & García, A. (2000). Seismovolcanic signals at deception island volcano, Antarctica: Wave field analysis and source modeling. *Journal of Geophysical Research*, *105*(B6), 13905–13931. https://doi.org/10.1029/2000JB900023

Iyer, H. (1992). Seismological detection and delineation of magma chambers: Present status with emphasis on the western USA. In P. Gasparini, R. Scarpa, & K. Aki (Eds.), *Volcanic Seismology* (Vol. 3, pp. 299–338). Springer. https://doi.org/10.1007/978-3-642-77008-1_22

Jousset, P., Currenti, G., Schwarz, B., Chalari, A., Tilmann, F., Reinsch, T., et al. (2022). Fibre optic distributed acoustic sensing of volcanic events. *Nature Communications*, *13*(1), 1753. https://doi.org/10.1038/s41467-022-29184-w

Jousset, P., Reinsch, T., Ryberg, T., Blanck, H., Clarke, A., Aghayev, R., et al. (2018). Fibre-optic distributed acoustic sensing data from Reykjanes, Iceland. *GFZ Data Services*. https://doi.org/10.5880/GFZ.6.2.2018.003

Kamath, U., Liu, J., & Whitaker, J. (2019). *Deep learning for NLP and speech recognition* (Vol. 84). Springer. https://doi.org/10.1007/978-3-030-14596-5

Klaasen, S., Paitz, P., Lindner, N., Dettmer, J., & Fichtner, A. (2021). Distributed acoustic sensing in volcano-glacial environments—Mount meager, British Columbia. *Journal of Geophysical Research: Solid Earth*, *126*(11), e2021JB022358. https://doi.org/10.1029/2021jb022358

Lahr, J. C., Chouet, B. A., Stephens, C. D., Power, J. A., & Page, R. A. (1994). Earthquake classification, location, and error analysis in a volcanic environment: Implications for the magmatic system of the 1989–1990 eruptions at redoubt volcano, Alaska. *Journal of Volcanology and Geothermal Research*, *62*(1–4), 137–151. https://doi.org/10.1016/0377-0273(94)90031-0

Lara, F., Lara-Cueva, R., Larco, J. C., Carrera, E. V., & León, R. (2021). A deep learning approach for automatic recognition of seismo-volcanic events at the Cotopaxi volcano. *Journal of Volcanology and Geothermal Research*, *409*, 107142. https://doi.org/10.1016/j.jvolgeores.2021.107142

Lara-Cueva, R., Benítez, D. S., Paillacho, V., Villalva, M., & Rojo-Álvarez, J. L. (2017). On the use of multi-class support vector machines for classification of seismic signals at Cotopaxi volcano. In *2017 IEEE International Autumn Meeting on Power, Electronics and Computing (ROPEC)* (pp. 1–6). https://doi.org/10.1109/ROPEC.2017.8215795

Lara-Cueva, R. A., Benítez, D. S., Carrera, E. V., Ruiz, M., & Rojo-Álvarez, J. L. (2016). Automatic recognition of long period events from volcano tectonic earthquakes at Cotopaxi volcano. *IEEE Transactions on Geoscience and Remote Sensing*, *54*(9), 5247–5257. https://doi.org/10.1109/TGRS.2016.2563894

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, *521*(7553), 436–444. https://doi.org/10.1038/nature14539

Li, J., Biondi, E., Heimisson, E. R., Puel, S., Zhai, Q., Zhang, S., et al. (2024). Minute-scale dynamics of recurrent dike intrusions in Iceland with fiber-optic geodesy. *Science*, *388*(6752), 1189–1193. https://doi.org/10.1126/science.adu0225

Lindsey, N. J., Martin, E. R., Dreger, D. S., Freifeld, B., Cole, S., James, S. R., et al. (2017). Fiber-optic network observations of earthquake wavefields. *Geophysical Research Letters*, *44*(23), 11–792. https://doi.org/10.1002/2017gl075722

Lindsey, N. J., Rademacher, H., & Ajo-Franklin, J. B. (2020). On the broadband instrument response of fiber-optic das arrays. *Journal of Geophysical Research: Solid Earth*, *125*(2), e2019JB018145. https://doi.org/10.1029/2019jb018145

Lior, I., Sladen, A., Rivet, D., Ampuero, J. P., Hello, Y., Becerril, C., et al. (2021). On the detection capabilities of underwater distributed acoustic sensing. *Journal of Geophysical Research: Solid Earth*, *126*(3), e2020JB020925. https://doi.org/10.1029/2020jb020925

Malfante, M., Dalla Mura, M., Mars, J. I., Métaxian, J. P., Macedo, O., & Inza, A. (2018). Automatic classification of volcano seismic signatures. *Journal of Geophysical Research: Solid Earth*, *123*(12), 10645. https://doi.org/10.1029/2018JB016370

Masotti, M., Falsaperla, S., Langer, H., Spampinato, S., & Campanini, R. (2006). Application of support vector machine to the classification of volcanic tremor at Etna, Italy. *Geophysical Research Letters*, *33*(20). https://doi.org/10.1029/2006GL027396

Münchmeyer, J., Bindi, D., Leser, U., & Tilmann, F. (2021). Earthquake magnitude and location estimation from real time seismic waveforms with a transformer network. *Geophysical Journal International*, *226*(2), 1086–1104. https://doi.org/10.1093/gji/ggab139

Nakano, M., Ichihara, M., Suetsugu, D., Ohminato, T., Ono, S., Vaiomounga, R., & Shinohara, M. (2024). Monitoring volcanic activity with distributed acoustic sensing using the Tongan seafloor telecommunications cable. *Earth Planets and Space*, *76*(1), 25. https://doi.org/10.1186/s40623-024-01972-2

Neuberg, J., Luckett, R., Ripepe, M., & Braun, T. (1994). Highlights from a seismic broadband array on Stromboli volcano. *Geophysical Research Letters*, *21*(9), 749–752. https://doi.org/10.1029/94GL00377

Nishimura, T., Emoto, K., Nakahara, H., Miura, S., Yamamoto, M., Sugimura, S., & Kimura, T. (2021). Source location of volcanic earthquakes and subsurface characterization using fiber-optic cable and distributed acoustic sensing system. *Scientific Reports*, *11*(1), 6319. https://doi.org/10.1038/s41598-021-85621-8

Pascanu, R., Mikolov, T., & Bengio, Y. (2013). On the difficulty of training recurrent neural networks. Retrieved from https://arxiv.org/abs/1211.5063

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., et al. (2019). Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32* (pp. 8024–8035). Curran Associates, Inc. Retrieved from. Retrieved from http://papers.neurips.cc/paper/9015-pytorch-an-imperative-style-high-performance-deep-learning-library.pdf

Ripepe, M., Donne, D. D., Harris, A., Marchetti, E., & Ulivieri, G. (2008). Dynamics of strombolian activity. In S. Calvari, S. Inguaggiato, G. Puglisi, M. Ripepe, & M. Rosi (Eds.), *The Stromboli Volcano: An Integrated Study of the 2002–2003 Eruption* (pp. 39–48). American Geophysical Union.

Roman, D. C., Rodgers, M., Geirsson, H., LaFemina, P. C., & Tenorio, V. (2016). Assessing the likelihood and magnitude of volcanic explosions based on seismic quiescence. *Earth and Planetary Science Letters*, *450*, 20–28. https://doi.org/10.1016/j.epsl.2016.06.020

Romero, J. E., Burton, M., Cáceres, F., Taddeucci, J., Civico, R., Ricci, T., et al. (2022). The initial phase of the 2021 Cumbre Vieja ridge eruption (Canary Islands): Products and dynamics controlling edifice growth and collapse. *Journal of Volcanology and Geothermal Research*, *431*, 107642. https://doi.org/10.1016/j.jvolgeores.2022.107642

Schmidhuber, J. (2015). Deep learning in neural networks: An overview. *Neural Networks*, *61*, 85–117. https://doi.org/10.1016/j.neunet.2014.09.003

Schwarz, B. (2019). An introduction to seismic diffraction. In *Advances in Geophysics* (Vol. 60, pp. 1–64). Elsevier. https://doi.org/10.1016/bs.agph.2019.05.001

Shang, Y., Sun, M., Wang, C., Yang, J., Du, Y., Yi, J., et al. (2022). Research progress in distributed acoustic sensing techniques. *Sensors*, *22*(16), 6060. https://doi.org/10.3390/s22166060

Small, C., & Naumann, T. (2001). The global distribution of human population and recent volcanism. Global Environmental Change Part B. *Environmental Hazards*, *3*(3), 93–109. https://doi.org/10.3763/ehaz.2001.0309

Titos, M., Bueno, A., García, L., & Benítez, C. (2018). A deep neural networks approach to automatic recognition systems for volcano-seismic events. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, *11*(5), 1533–1544. https://doi.org/10.1109/JSTARS.2018.2833145

Titos, M., Bueno, A., García, L., Benítez, C., & Segura, J. C. (2019). Classification of isolated volcano-seismic events based on inductive transfer learning. *IEEE Geoscience and Remote Sensing Letters*, *17*(5), 869–873. https://doi.org/10.1109/LGRS.2019.2903053

Titos, M., Bueno, A., García, L., Benítez, M. C., & Ibañez, J. (2018). Detection and classification of continuous volcano-seismic signals with recurrent neural networks. *IEEE Transactions on Geoscience and Remote Sensing*, *57*(4), 1936–1948.

Titos, M., Carthy, J., García, L., Barnie, T., & Benítez, C. (2024). Dilated-RNNs: A deep approach for continuous volcano-seismic events recognition. *Ieee Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, *17*, 11857–11865. https://doi.org/10.1109/JSTARS.2024.3421921

Torres-González, P. A., Luengo-Oroz, N., Lamolda, H., D'Alessandro, W., Albert, H., Iribarren, I., & Soler, V. (2020). Unrest signals after 46 years of quiescence at Cumbre Vieja, La Palma, Canary Islands. *Journal of Volcanology and Geothermal Research*, *392*, 106757. https://doi.org/10.1016/j.jvolgeores.2019.106757

Trnkoczy, A. (2009). Understanding and parameter setting of STA/LTA trigger algorithm. In *New Manual of seismological Observatory Practice (NMSOP)* (pp. 1–20). Deutsches GeoForschungsZentrum GFZ.

Troll, V. R., Aulinas, M., Carracedo, J. C., Geiger, H., Perez-Torrado, F. J., Soler, V., et al. (2024). The 2021 La Palma eruption: Social dilemmas resulting from life close to an active volcano. *Geology Today*, *40*(3), 96–111. https://doi.org/10.1111/gto.12472

Walter, F., Gräff, D., Lindner, F., Paitz, P., Köpfli, M., Chmiel, M., & Fichtner, A. (2020). Distributed acoustic sensing of microseismic sources and wave propagation in glaciated terrain. *Nature Communications*, *11*(1), 2436. https://doi.org/10.1038/s41467-020-15824-6

Wassermann, J., Braun, T., Ripepe, M., Bernauer, F., Guattari, F., & Igel, H. (2022). The use of 6DOF measurement in volcano seismology–a first application to Stromboli volcano. *Journal of Volcanology and Geothermal Research*, *424*, 107499. https://doi.org/10.1016/j.jvolgeores.2022.107499

White, R., & McCausland, W. (2016). Volcano-tectonic earthquakes: A new tool for estimating intrusive volumes and forecasting eruptions. *Journal of Volcanology and Geothermal Research*, *309*, 139–155. https://doi.org/10.1016/j.jvolgeores.2015.10.020

Williams, E., Fernández-Ruiz, M. R., Magalhães, R., Vanthillo, R., Zhan, Z., Gonzalez-Herraez, M., & Martins, H. (2019). Teleseisms and microseisms on an ocean-bottom distributed acoustic sensing array. https://doi.org/10.31223/osf.io/kg7q4

Xu, R., Wunsch, D., II., & Frank, R. (2007). Inference of genetic regulatory networks with recurrent neural network models using particle swarm optimization. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, *4*(4), 681–692. https://doi.org/10.1109/tcbb.2007.1057

Zhan, Z. (2020). Distributed acoustic sensing turns fiber-optic cables into sensitive seismic antennas. *Seismological Research Letters*, *91*(1), 1–15. https://doi.org/10.1785/0220190112

Zhu, W., Biondi, E., Li, J., Yin, J., Ross, Z. E., & Zhan, Z. (2023a). Seismic arrival-time picking on distributed acoustic sensing data using semi-supervised learning. *Nature Communications*, *14*(1), 8192. https://doi.org/10.1038/s41467-023-43355-3

Zhu, W., Biondi, E., Li, J., Yin, J., Ross, Z. E., & Zhan, Z. (2023b). Seismic arrival-time picking on distributed acoustic sensing data using semi-supervised learning [Software]. *Zenodo*. https://doi.org/10.5281/zenodo.10043122

Zobin, V. M. (2003). *Introduction to volcanic seismology*. Elsevier. https://doi.org/10.1016/C2011-0-06141-0